

DSA0502 – Query Processing for Data Science	
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CO1 AT2 Application-Based Questions

Question 1

Employee Data Cleaning (CSV)

A company stores employee information in employees.csv. Some employee names contain extra spaces, salaries are stored as text, and a few records have missing values.

Task

- Read the CSV file.
- Remove extra spaces from employee names.
- Convert salary into numeric format.
- Fill missing salary values with the average salary.
- Save the cleaned dataset as cleaned_employees.csv.

Aim

To perform employee data cleaning using Pandas by reading a CSV file, removing unnecessary spaces from employee names, converting salary values into numeric format, filling missing salary values with the average salary, and exporting the cleaned dataset.

Dataset (Save as employees.csv)

```
%%writefile employees.csv
```

```
EmployeeID,Name,Department,Salary
```

```
E101," John ",HR,50000
```

```
E102,"Alice ",Finance,60000
```

```
E103," David",IT,
```

```
E105," Sophia ",HR,55000
```

```
E106,"Michael ",Finance,
```

```
E107," Daniel",IT,65000
```

```
E108,"Olivia ",Marketing,58000
```

Program

```
import pandas as pd

df = pd.read_csv("employees.csv")

print("Original Dataset:")

print(df)

df["Name"] = df["Name"].str.strip()

df["Name"] = df["Name"].str.replace(r"\s+", " ", regex=True)

# Convert Salary column to numeric

df["Salary"] = pd.to_numeric(df["Salary"], errors="coerce")

# Fill missing salary values with average salary

average_salary = df["Salary"].mean()

df["Salary"] = df["Salary"].fillna(average_salary)

print("\nAverage Salary:", average_salary)

print("\nCleaned Dataset:")

print(df)

df.to_csv("cleaned_employees.csv", index=False)

print("\nFile saved successfully as 'cleaned_employees.csv'")
```

Output

```
... Original Dataset:
  EmployeeID  Name Department  Salary
0      E101   John         HR   50000.0
1      E102  Alice         Finance 60000.0
2      E103   David          IT      NaN
3      E104   Emma      Marketing 70000.0
4      E105  Sophia          HR   55000.0
5      E106 Michael         Finance      NaN
6      E107   Daniel          IT   65000.0
7      E108  Olivia      Marketing 58000.0

Average Salary: 59666.666666666664

Cleaned Dataset:
  EmployeeID  Name Department  Salary
0      E101   John         HR  50000.000000
1      E102  Alice         Finance 60000.000000
2      E103   David          IT  59666.666667
3      E104   Emma      Marketing 70000.000000
4      E105  Sophia          HR  55000.000000
5      E106 Michael         Finance 59666.666667
6      E107   Daniel          IT  65000.000000
7      E108  Olivia      Marketing 58000.000000

File saved successfully as 'cleaned_employees.csv'
```

Result

The employee dataset was successfully cleaned by removing extra spaces from employee names, converting salary values into numeric format, replacing missing salary values with the average salary, and saving the cleaned data as cleaned_employees.csv.

Question 2: Student Marks Analysis

A college maintains student marks in `students.csv`.

Task:

- Read the dataset.
- Calculate the average marks.
- Identify students who scored below 40.
- Add a new column named **Result (Pass/Fail)**.
- Export the updated file.

Aim

To read a student marks dataset, calculate the average marks, identify students scoring below 40, add a Result (Pass/Fail) column, and export the updated dataset.

Dataset (`students.csv`)

```
%%writefile students.csv
```

```
StudentID,Name,Marks
```

```
S101,Alice,85
```

```
S102,Bob,35
```

```
S103,Charlie,78
```

```
S104,David,25
```

```
S105,Emma,90
```

```
S106,Frank,40
```

```
S107,Grace,55
```

```
S108,Henry,30
```

Python Code

```
import pandas as pd
```

```
# Read the CSV file
```

```
df = pd.read_csv("students.csv")
```

```
print("Original Dataset:")
```

```
print(df)
```

```
# Calculate average marks
```

```
average_marks = df["Marks"].mean()
```

```
print("\nAverage Marks:", average_marks)
```

```

# Display students who scored below 40

print("\nStudents who scored below 40:")

print(df[df["Marks"] < 40])

# Add Result column (Pass/Fail)

df["Result"] = df["Marks"].apply(lambda x: "Pass" if x >= 40 else "Fail")

print("\nUpdated Dataset:")

print(df)

# Save the updated dataset

df.to_csv("updated_students.csv", index=False)

print("\nFile saved successfully as 'updated_students.csv'")

```

Output

```

Original Dataset:
  StudentID  Name  Marks
0    S101  Alice    85
1    S102   Bob    35
2    S103 Charlie    78
3    S104  David    25
4    S105  Emma    90
5    S106  Frank    40
6    S107  Grace    55
7    S108  Henry    30

Average Marks: 54.75

Students who scored below 40:
  StudentID  Name  Marks
1    S102   Bob    35
3    S104  David    25
7    S108  Henry    30

Updated Dataset:
  StudentID  Name  Marks  Result
0    S101  Alice    85    Pass
1    S102   Bob    35    Fail
2    S103 Charlie    78    Pass
3    S104  David    25    Fail
4    S105  Emma    90    Pass
5    S106  Frank    40    Pass
6    S107  Grace    55    Pass
7    S108  Henry    30    Fail

File saved successfully as 'updated_students.csv'

```

Result

The student marks dataset was successfully analyzed by calculating the average marks, identifying students who scored below 40, adding a **Result** column indicating Pass or Fail, and exporting the updated dataset as **updated_students.csv**.